# Report: Storyboarding and User Interface Design Modeling for MomentsHub

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**Project:** MomentsHub: A Collaborative Daily Photo Capsule

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### 1. Introduction

In the development of any software, a critical gap exists between the formal, technical specifications of a system and the fluid, intuitive experience desired by the end-user. For the MomentsHub project, this is the gap between the functional requirements (FRs) listed in the Software Requirements Specification (SRS) and the goal of creating an "intimate space for friends and family".

To bridge this gap, two design techniques are essential: **User Interface (UI) Design Modeling** and **Storyboarding**.

1. **UI Design Modeling** provides the visual and interactive blueprint of the application—the *what* and *how* of the user's interaction.
2. **Storyboarding** provides the narrative context—the *who*, *where*, and *why*—that drives the user's interaction and validates the application's flow.

This report will define both techniques and present a 5-frame storyboard for a core function of the MomentsHub application: the daily photo upload.

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### 2. User Interface (UI) Design Modeling

UI Design Modeling is the iterative process of creating visual representations of the application's interface. These models evolve from low-fidelity (basic) to high-fidelity (detailed), allowing the development team to test and refine ideas before writing code.

#### 2.1. Levels of Fidelity

- **Low-Fidelity (Wireframes):**
    - **Description:** These are simple, black-and-white layouts that focus on structure, space allocation, and functional flow. They use basic shapes (boxes, lines, placeholder text) to represent UI elements.
    - **Application to MomentsHub:** A wireframe for the MomentsHub dashboard would simply be a large box labeled "On This Day Section" , a grid of smaller boxes for the "Group Gallery View", and a circle icon

for the "Upload Photo" button. The goal is to agree on *layout*, not aesthetics.

- **High-Fidelity (Mockups):**
    - **Description:** These are static, detailed, and visually rich representations of the final product. They include the chosen color scheme, typography, iconography, and branding.
    - **Application to MomentsHub:** A mockup would be a full-color, realistic image of the dashboard, styled using Tailwind CSS. It would show the exact font, the five predefined emoji reactions , and the final look of the React components.
- **Prototypes:**
    - **Description:** These are interactive, clickable mockups. They simulate the application's flow, allowing a user to click a button and be taken to the next screen.
    - **Application to MomentsHub:** A prototype would allow a stakeholder (like your faculty guide ) to click the "Upload Photo" button, which would then *actually* open the uploadForm: UploadFormUI screen, simulating the flow shown in your Sequence Diagram.

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### 3. Storyboarding

Storyboarding is a narrative technique that visually details a user's journey as they interact with the application to achieve a specific goal. It is essentially a comic strip for a user scenario.

#### 3.1. Purpose and Benefits

- **Builds Empathy:** It forces the developer to think from the user's perspective, considering their thoughts, feelings, and real-world environment.
- **Provides Context:** A requirement like "A user shall be able to upload one... photo per day" is a *what*. A storyboard shows *why*—a user sees a beautiful scene and *feels* a connection to their family.
- **Validates Flow:** It tests the logic of your diagrams. The flow you defined in the Activity Diagram and State Diagram can be walked through in a real-world scenario to see if it feels natural.
- **Identifies Gaps:** It reveals missing steps. The storyboard we built identified the need to show a "success" message, a small but vital piece of user feedback.

#### 3.2. Connection to MomentsHub Artifacts

- SRS : A storyboard is a visualization of a "Stimulus/Response Sequence" . The user seeing a flower is the "Stimulus," and their action of opening the app is the "Response."
- Use Case Diagram : It provides a detailed, human-centric narrative for a single Use Case, such as "Upload Photo".
- Sequence Diagram : The storyboard provides the human story *before* and *after* the technical sequence. The user's goal (Frame 1-3) *triggers* the sequence (Frame 4), and the user's satisfied reaction (Frame 5) is the *result* of that sequence.

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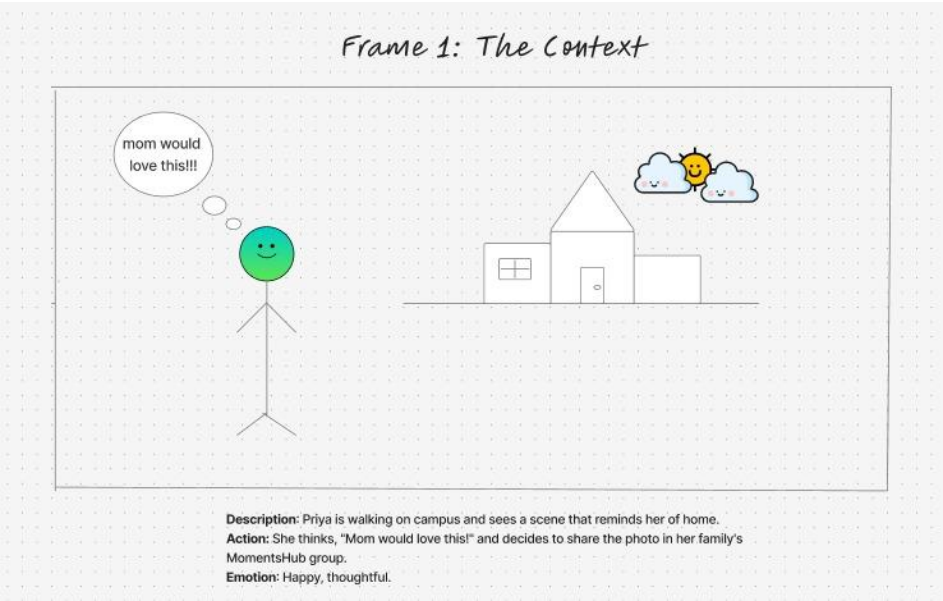
**4. Storyboard for MomentsHub: "Priya's Daily Photo"**

This storyboard, based on the frames you created, details the core "Upload Photo" functionality.

- **Persona:** Priya (a Group Member )
- **Scenario:** Priya is a college student in a MomentsHub group with her family, who live in another city. While walking on campus, she sees a scene that reminds her of home.
- **Goal:** To upload her daily photo and share the moment with her family.

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**Frame 1: The Context**

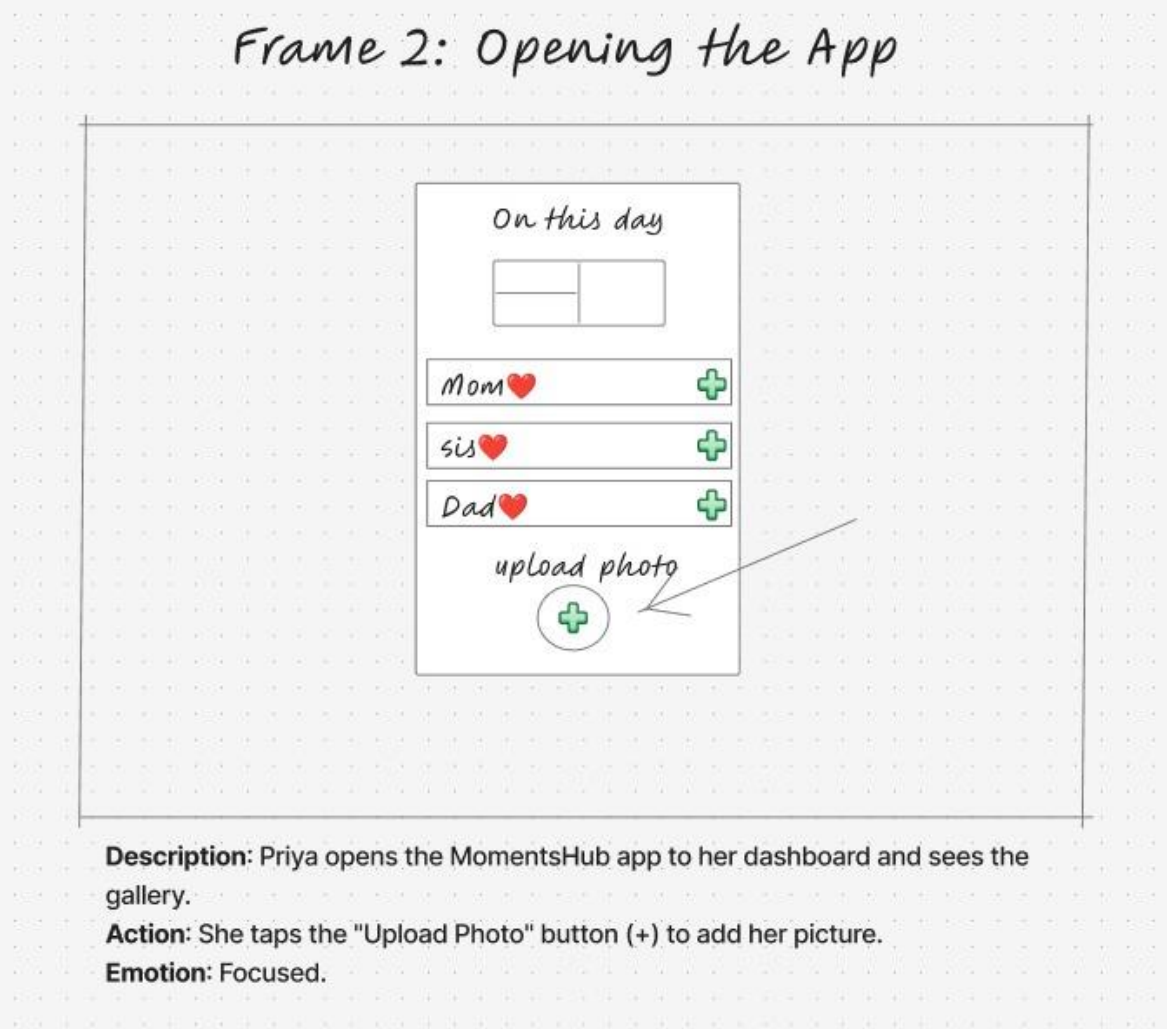



- **Visual:** Priya (stick figure) is outside on campus. A house and sun are visible. A thought bubble says, "mom would love this!!!"

- **Description:** Priya is walking on campus and sees a scene that reminds her of home.
- **Action:** She thinks, "Mom would love this!" and decides to share the photo in her family's MomentsHub group.
- **Emotion:** Happy, thoughtful.

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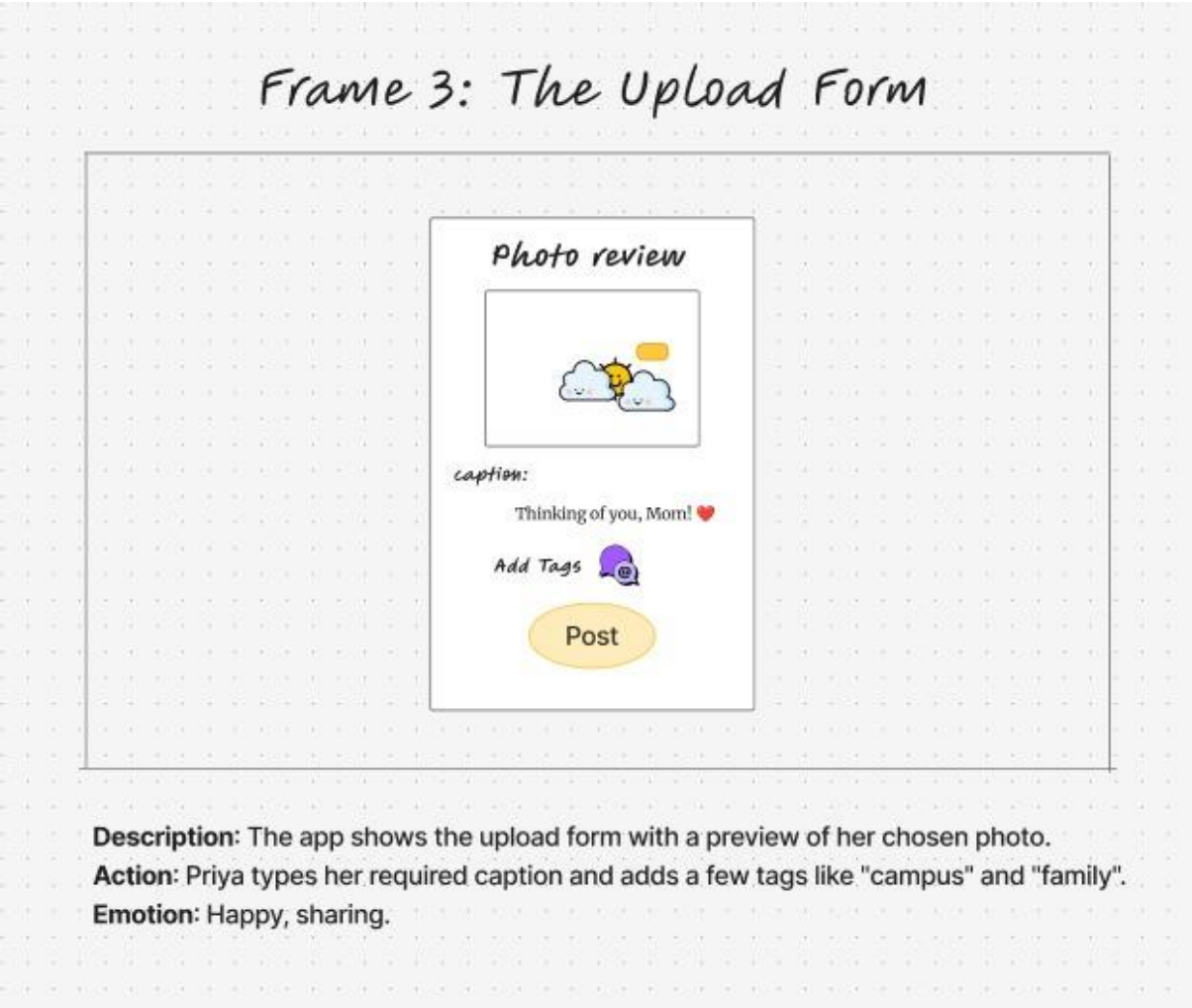
**Frame 2: Opening the App**




- **Visual:** A phone screen shows the MomentsHub dashboard. It displays "On this day" and a list of group members. A finger points to the main "upload photo" + button at the bottom.
- **Description:** Priya opens the MomentsHub app to her dashboard and sees the gallery.
- **Action:** She taps the "Upload Photo" button (+) to add her picture.
- **Emotion:** Focused.

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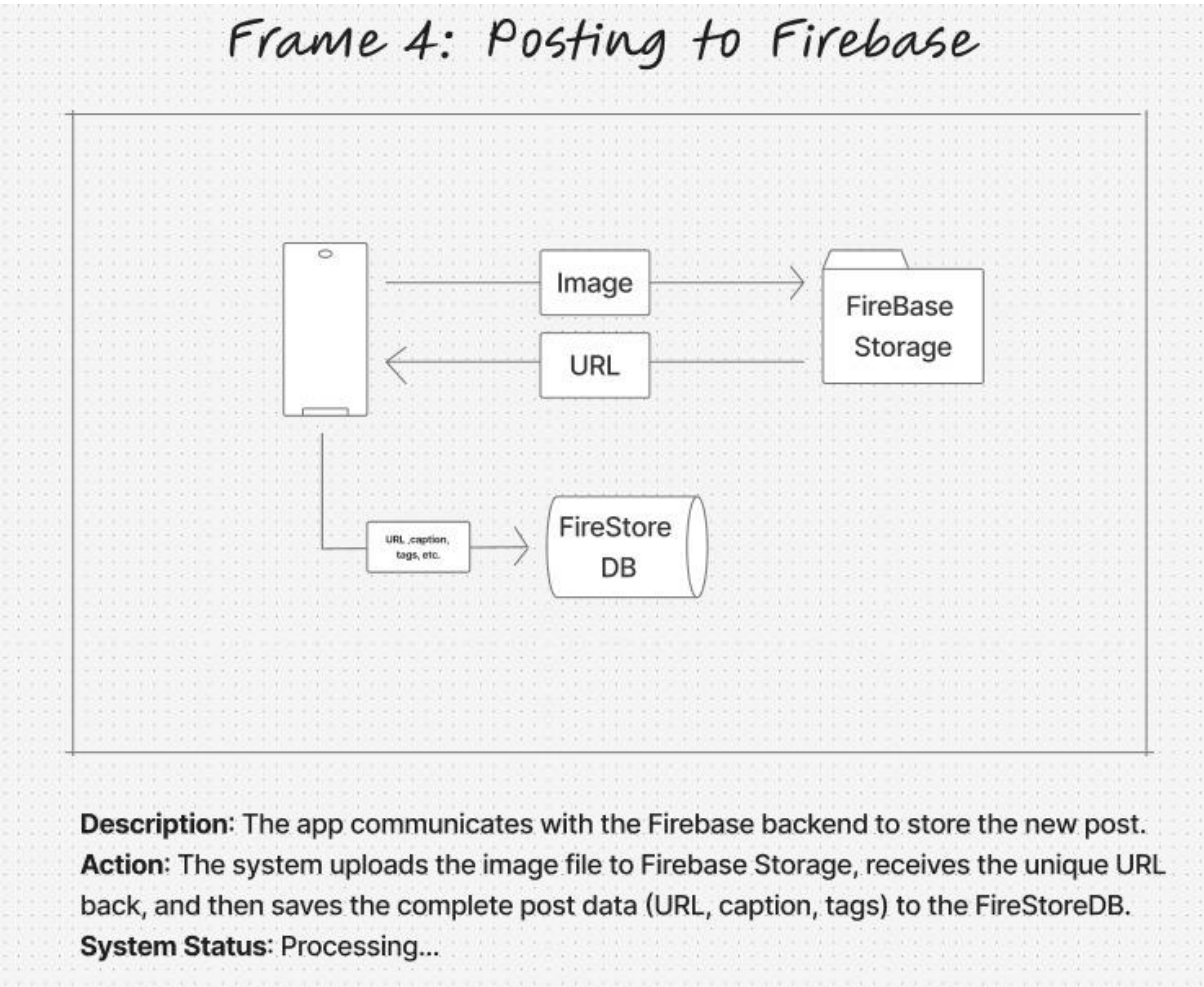
**Frame 3: The Upload Form**




- **Visual:** The phone screen shows the UploadFormUI. A "Photo review" preview is at the top. Below, a caption is filled in ("Thinking of you, Mom! ♡"), and a tag (campus) is added. A finger points to the "Post" button.
- **Description:** The app shows the upload form with a preview of her chosen photo.
- **Action:** Priya types her required caption and adds a few tags like "campus" and "family".
- **Emotion:** Happy, sharing.

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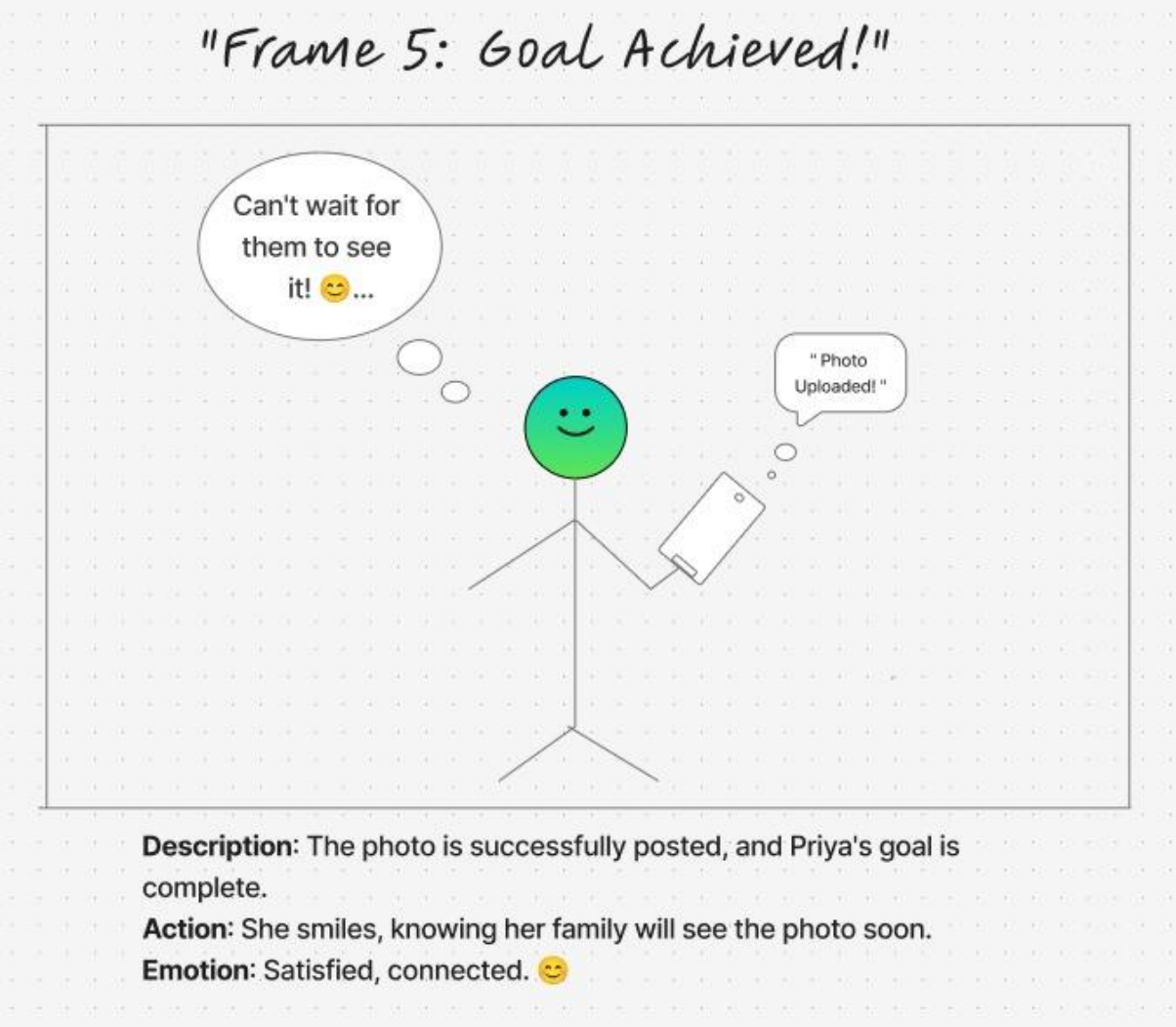
**Frame 4: Posting to Firebase**




- **Visual:** A technical diagram showing the Phone sending an Image to FireBase Storage. FireBase Storage returns a URL to the Phone. The Phone then sends the URL, caption, tags, etc. to the FireStoreDB cylinder.
- **Description:** The app communicates with the Firebase backend to store the new post.
- **Action:** The system uploads the image file to Firebase Storage, receives the unique URL back, and then saves the complete post data (URL, caption) to the FireStoreDB.
- **System Status:** Processing...

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**Frame 5: Goal Achieved!**




- **Visual:** Priya (stick figure) is smiling and holding her phone. A thought bubble from the phone says, "Photo Uploaded!".
- **Description:** The photo is successfully posted, and Priya's goal is complete.
- **Action:** She smiles, knowing her family will see the photo soon.
- **Emotion:** Satisfied, connected. 😊

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**5. Conclusion**

Storyboarding and UI Design Modeling are indispensable techniques for the MomentsHub project. They act as the essential bridge between the technical requirements of the SRS and the human-centered goal of fostering genuine connection.

By first modeling the interface with wireframes and mockups, and then validating its flow with a context-rich storyboard, the development team can ensure the application is logical, intuitive, and emotionally resonant. This process de-risks development, clarifies implementation logic, and ultimately leads to a product that users will find both functional and meaningful.